

Assignment 3: Recent Architectural and Regularization Developments in Semi-Parametric Generalized Additive Models

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1. CONTEXT AND LITERATURE SELECTION

Generalized Additive Models (GAMs) occupy a distinct niche in intermediate statistical learning, offering a flexible framework that circumvents the rigid linearity of classical GLMs without collapsing into the absolute lack of interpretability typical of deep neural networks or ensemble trees. Despite their structural utility, traditional backfitting estimation for GAMs struggles when confronted with massive feature spaces or irregular data topologies. This review contextualizes these classical limits against two modern methodological adaptations: the Neural Additive Model (NAM) paradigm detailed by Agarwal et al. (2021) and contemporary sparse regularization methodologies for missing data structures within additive layers (M-GAM, 2024).

The core theme tying these papers together is the effort to retain the underlying additive interpretability of the model while altering the computational engine. Agarwal et al. substitute standard spline smoother functions with isolated, single-input neural networks to leverage gradient descent optimization. Conversely, the M-GAM framework expands the structural capabilities of additive models by formulating sparsity constraints directly over missing data indicators, optimizing the shape functions without requiring pre-modeling feature imputation.

2. FORMAL MODELING ARCHITECTURE

The mathematical foundation of a classical GAM generalizes standard regression by mapping an expected response to an additive sequence of smooth, non-parametric functions. Formally, for a target variable Y and a collection of p predictors x , the structural link is written as:

$$g(E[Y]) = \beta_0 + \sum_{j=1}^p f_j(x_j)$$

where $g(\cdot)$ represents a monotonic link function, β_0 is the intercept, and each $f_j(\cdot)$ is an arbitrary univariate smoother optimized via penalized splines or local scoring.

The Neural Additive Model fundamentally changes this estimation pipeline by re-interpreting each smoother as a separate parameter block. The updated setup is formalized as:

$$g(E[Y]) = \beta_0 + \sum_{j=1}^p NN_j(x_j)$$

Here, each $NN_j(\cdot)$ is a completely independent feedforward neural network that processes only the scalar input x_j . The crucial architectural constraint is the absolute absence of cross-talk or hidden layer interleaving between different features. Because the feature representations remain isolated until they are linearly combined at the final output layer, the configuration strictly preserves the additive identity of the model, allowing the global impact of a single covariate to be isolated and evaluated independently of others.

3. METHODOLOGICAL BREAKTHROUGHS

Standard multi-layer perceptrons generally perform poorly on tabular fields because continuous hidden layers tend to over-smooth sharp, abrupt functional adjustments. To resolve this specific limitation, Agarwal et al. developed the Exponential Unit (ExU) activation function. The structural activation within the base layers of each single-feature network is defined as:

$$h(x) = f(e^w \cdot (x - b))$$

By forcing the weight parameter w into an exponential space (e^w), the network forces its hidden states to be highly sensitive to small variations in the input space. This allows the individual neural blocks to accurately fit highly irregular, jagged, or near-discontinuous trajectories, directly matching the performance of gradient-boosted trees while maintaining smooth differentiation boundaries.

Parallel to this architectural shift, recent work in handling unobserved data within semi-parametric frameworks (M-GAM) introduced an ℓ_0 -norm regularized formulation directly targeting missingness paths. Instead of relying on a separate imputation process that often distorts the curvature of f_j , the missing indicators are built directly into the penalty function. This allows the model to map missing entries as discrete structural offsets, isolating the optimization of the smooth functions exclusively to valid data intervals.

4. CURRICULAR RELEVANCE AND EMPIRICAL APPLICATIONS

These developments tie directly into core statistical learning themes, specifically the bias-variance trade-off and computational scaling under non-parametric penalties. Traditional backfitting algorithms operate via iterative coordinate descent, a process that is highly sequential and computationally inefficient for high-throughput data. Shifting the optimization to a neural additive framework allows the model to be trained using standard mini-batch stochastic gradient descent (SGD) and backpropagation. This enables full integration with modern deep learning hardware configurations and regularization techniques like dropout and weight decay.

In practice, these modern models are crucial for risk-sensitive deployments like medical diagnostics and credit assessment pipelines. In financial compliance, machine learning models must provide clear justification for automated rejections. While standard neural networks fail this constraint due to internal feature entanglement, a neural GAM allows operators to extract and plot the exact risk

curve of each metric, providing an auditable model that matches the predictive performance of a black-box ensemble.

5. CRITICAL EVALUATION

While utilizing deep architectures within classical statistical frameworks is elegant, it introduces several distinct practical liabilities. The assumption of total additivity is fundamentally a structural compromise. In real-world environments, complex cross-feature interactions are common. Extending these models to capture pairwise associations requires adding explicit bivariate sub-networks, which triggers an exponential expansion of parameters that quickly degrades both computational efficiency and visual clarity.

Additionally, optimizing a large collection of disjoint networks introduces significant variance across different random seeds. When features exhibit severe multi-collinearity, the gradient updates can become unstable, leading to situations where variance is arbitrarily shifted between competing sub-networks. This instability can result in conflicting shape curves across separate training runs, indicating that while neural GAMs improve scalability, they require careful regularization to ensure stable interpretation.

REFERENCES

- Agarwal, R., Melnick, L., Frosst, N., Zhang, X., Lengerich, B., Caruana, R., & Hinton, G. E. (2021). Neural Additive Models: Interpretable machine learning with neural nets. *Advances in Neural Information Processing Systems*, 34, 4699-4711.
- Hastie, T., & Tibshirani, R. (1990). *Generalized Additive Models*. London: Chapman and Hall.
- Lou, Y., Caruana, R., Gehrke, J., & Giles, C. L. (2013). Accurate intelligible models with pairwise interactions. *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 623-631.